

Crosstabs & Contingency Analysis

Method Dossier with Field-Specific Examples | Entry Level

1. Method Description

Imagine: You surveyed 200 people about their gender (m/f/d) and whether they exercise regularly (yes/no). The raw data is a long list of individual answers. To get an overview, you need to sort the data into a table. That is exactly what a crosstab (contingency table) does: it counts how many people have each combination of two categorical variables.

Crosstabs are the first step in any categorical data analysis. They show immediately whether there might be a relationship between two variables — even before running a chi-square test.

Structure of a Crosstab

A crosstab consists of:

- Rows: categories of the first variable (e.g., gender)
- Columns: categories of the second variable (e.g., exercise)
- Cells: number of cases per combination (absolute frequency)
- Margins: row and column totals
- Grand total: N (all cases)

Example 1: Crosstab Gender × Exercise (absolute frequencies)

	Exercise: Yes	Exercise: No	Total
Male	60	40	100
Female	75	25	100
Diverse	10	10	20
Total	145	75	220

Row Percentages, Column Percentages, Total Percentages

The key insight: Which percentage you use depends on your research question.

ROW PERCENTAGES (→): Each row sums to 100%. You compare the distribution of the column variable WITHIN each row. Use when the row variable is the independent variable (IV).

Question: 'Does the exercise rate differ between genders?' Row percentages: 60% exercise among men, 75% among women, 50% among diverse.

Example 2: Row percentages — comparison between groups (IV = Gender)

	Exercise: Yes	Exercise: No	Total
Male	60 %	40 %	100 %
Female	75 %	25 %	100 %
Diverse	50 %	50 %	100 %

COLUMN PERCENTAGES (↓): Each column sums to 100%. You compare the distribution of the row variable WITHIN each column. Use when the column variable is the IV.

Question: 'How is gender distributed among exercisers vs. non-exercisers?'

Example 3: Column percentages — comparison between columns (IV = Exercise)

	Exercise: Yes	Exercise: No
Male	41.4 %	53.3 %
Female	51.7 %	33.3 %
Diverse	6.9 %	13.3 %
Total	100 %	100 %

TOTAL PERCENTAGES: Each cell is divided by the grand total N. You see which combination is most frequent overall. Useful for purely descriptive overviews without causal direction.

Question: 'Which group is most represented overall?' 34.1% = female + exercise.

Example 4: Total percentages — all cells relative to N

	Exercise: Yes	Exercise: No	Total
Male	27.3 %	18.2 %	45.5 %
Female	34.1 %	11.4 %	45.5 %
Diverse	4.5 %	4.5 %	9.0 %
Total	65.9 %	34.1 %	100 %

Rule of thumb: Which percentage to use?

- IV in rows, DV in columns row percentages (compare DV across groups)
- IV in columns, DV in rows column percentages
- Exploratory overview without causal direction total percentages
- Absolute frequencies alone insufficient for unequal group sizes

2. Field-Specific Examples

Social Sciences

Question: Does education level (secondary/A-levels/university) correlate with voting preference (Party A/B/C)? Hypothesis: Education influences voting. Data source: ALLBUS / SOEP. Method: Crosstab with row percentages + chi-square test.

Economics / Business

Question: Is there a relationship between company size (SME/large) and digitalization level (low/medium/high)? Data: IW survey. Interpretation: Row percentages show large companies are more often 'highly' digitalized.

Health Sciences / Medicine

Question: Smoking (yes/no) × lung cancer diagnosis (yes/no). Study with $n = 500$. 2×2 table. Row percentages: smokers have higher cancer rate. Extension: calculating relative risk (RR) and odds ratio (OR).

Humanities

Question: Corpus linguistics — use of 'der/die/das' in 18th vs. 20th century texts (masculine/feminine/neuter per period). Crosstab: Period × Article. Row percentages show the shift in article usage over time.

Sports Sciences

Question: Training method (A/B/C) × injury status (injured/not injured). $n = 90$ (30 per method). Crosstab + column percentages: which method has the lowest injury rate?

3. Assumptions and Prerequisites

- Both variables must be categorical (nominal or ordinal scale)
- Independence of observations (each person counted only once)
- Expected cell frequencies ≥ 5 for chi-square (otherwise: Fisher's exact test)
- No empty categories (remove unobserved levels)
- Always state the percentage base (e.g., 'row %', $n = 100$)

4. Interpretation and Reporting

Always report a crosstab as follows:

- Table with absolute frequencies + chosen percentage type
- State row/column totals and grand total N
- Describe patterns: 'In group X, the proportion of Y is Z %, higher/lower than in group W'
- Append chi-square test result if formally testing the relationship

Report example:

'Table 1 shows the crosstab of gender and regular exercise ($n = 220$). The proportion of exercisers is higher among women (75 %) than among men (60 %) and diverse individuals (50 %). A chi-square test confirms a significant relationship ($\chi^2(2) = 8.24$, $p = 0.016$, Cramér's $V = 0.19$).'

5. Code Examples

R — Crosstab + Percentages:

```
# Create crosstab
tab <- table(d$gender, d$exercise)
tab

# Row percentages
prop.table(tab, margin = 1) * 100 # rows = 1

# Column percentages
prop.table(tab, margin = 2) * 100 # columns = 2

# Total percentages
prop.table(tab) * 100

# With vcd package (nicer output)
library(vcd)
addmargins(tab) # with margins

# Chi-square test
chisq.test(tab)
```

Python (Pandas + SciPy) — Crosstab + Percentages:

```
import pandas as pd
from scipy.stats import chi2_contingency

# Create crosstab
tab = pd.crosstab(d['gender'], d['exercise'])
print(tab)

# Row percentages
print(tab.div(tab.sum(axis=1), axis=0) * 100)

# Column percentages
print(tab.div(tab.sum(axis=0), axis=1) * 100)

# Total percentages
print(tab / tab.values.sum() * 100)

# Chi-square test
chi2, p, dof, expected = chi2_contingency(tab)
print(f' $\chi^2 = \{chi2:.2f\}$ , p =  $\{p:.4f\}$ ')
```

SPSS — Crosstab + Percentages:

```
* Crosstab with row percentages
CROSSTABS
/TABLES=gender BY exercise
/FORMAT=AVALUE TABLES
/CELLS=COUNT ROW COLUMN TOTAL
/STATISTICS=CHISQ PHI.
```

```
* Output:
* COUNT = absolute frequencies
* ROW = row percentages
* COLUMN = column percentages
* TOTAL = total percentages
* CHISQ = chi-square test
* PHI = Cramér's V / Phi
```

6. Extension: Simpson's Paradox

A key pitfall with crosstabs: the apparent relationship between two variables can reverse when a third (confounding) variable is included. This is Simpson's Paradox.

Classic example: Gender bias in UC Berkeley admissions. Overall, men are admitted at a higher rate than women. However, within each department, the ratio is balanced. The reason: women applied more often to departments with lower admission rates.

Lesson: A single 2x2 crosstab can be misleading. When suspecting a confounder, use stratified tables (Mantel-Haenszel test) or logistic regression.

7. Exercises

Exercise 1: A sports club surveys 150 members on training type (endurance/strength/combined) and injury (yes/no). Create a crosstab and calculate row percentages. Which training type has the lowest injury rate?

Exercise 2: A survey (n = 400) examines the relationship between field of study (STEM/Languages/Social Sciences) and study abroad (yes/no). Calculate column percentages. Are the differences significant?

Exercise 3 (Simpson's Paradox): Two hospitals (A and B) each treat 100 patients with either the old or new method. In both hospitals, the new method is superior, but pooled together the old method appears better. Why? Consider: how do the hospitals differ in their case composition?